



Fusing Multi-scale Attention and Transformer for Detection and Localization of Image Splicing Forgery

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Abstract. Image forgery detection has attracted widespread attention due to the enormous spread of forged images on the internet and social media. Many existing methods lack global contextual information and ignore the interaction between multi-level representations, which are not conducive to localizing the multi-scale tampered regions. To overcome these limitations, we present a novel method to detect splicing forgery of images by utilizing cross-scale interaction attention and multi-level global information. Firstly, we design a cross-scale interactive attention (CSIA) module for aggregating multi-level convolutional features, focusing selectively on task-relevant information. It allows representations learned at different levels to communicate effectively with each other. Secondly, by introducing the transformer layers with dynamic position embedding, the proposed method can capture the multi-level global contextual correlations of the image in a more flexible way. Extensive experiments have shown that our proposed method outperforms state-of-the-art methods. It can effectively detect, and segment multi-scale tampered regions by aggregating multi-scale local and global feature relevance.

Keywords: Image forgery · Cross-scale interaction attention · Transformer · Global contextual information

1 Introduction

Image forgery is the use of image processing software to manipulate the semantic content of an image to distort the original information conveyed by the image. Splicing and copy-move are two common methods of tampering in real life. Splicing forgery selects random regions from donor images and pastes them into the target image for image composition. In splicing forgery, the forged and real regions are from different source images, as shown in Fig. 1-(a). Due to various post-processing operations masking the traces of manipulation, the naked eye can barely distinguish between the real and tampered areas of the image. Moreover, the number and scale of the forged areas contained in the splicing forged images are different. These factors make the detection of splicing forgery regions a challenging task.

The tampered images are widely spread on the media and internet. To avoid being tricked by malicious manipulated images, many forgery detection methods have been devised over the past decades. These methods can be broadly classified into two classes: conventional methods and deep learning-based methods. The conventional methods mainly utilize local inconsistencies in specific attributes to locate tampered regions, such as image essence attributes [1], compression artifacts [2], and photo-response non-uniformity (PRNU) [6]. The experimental results of the DCT [11] are shown in Fig. 1-(c). In recent years, inspired by the success of deep learning in vision tasks, some researchers have exploited CNNs to extract features from image patches and locate tampered regions by classifying patches [7, 8]. However, these methods lack contextual information about the image. To solve this problem, end-to-end models [9, 10] have been proposed. The experimental results of the ManTra [10] are shown in Fig. 1-(d). These models typically extract features from the entire image without differentiating the importance of the information. Moreover, most of these methods rely only on single-level local features without considering the multi-level local and global information of the image. Such factors hinder these methods from achieving state-of-the-art accuracy when detecting multi-scale splicing forgeries.

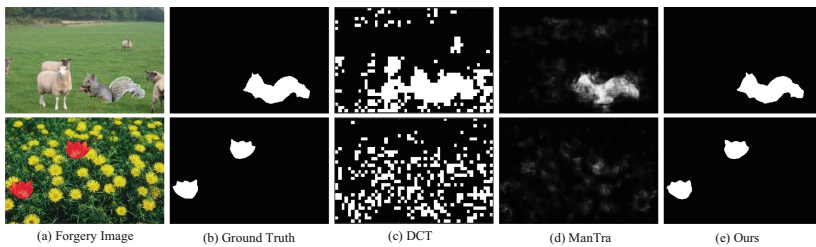


Fig. 1. Two examples of splicing forgery images and corresponding localization results of three methods. (a) The splicing forgery images. (b) Ground truths. (c) The detection results of the traditional method of DCT [11]. (d) The detection result of the deep learning method ManTra [10]. (e) The detection result of the proposed method.

To solve the issues mentioned above, we propose a fusing multi-scale attention and transformer network (FMAT-Net) for the detection of splicing forgery images by aggregating the multi-scale attention and global contextual information. This study makes the following contributions:

- We propose a novel method for the detection and localization of image splicing forgery. It can improve the localization performance of the forged regions by integrating multi-level local attention and global contextual information.
- The designed CSIA module allows adequate interaction of features at different levels and can enhance forgery-related features and suppress irrelevant responses. By introducing the transformer layers with convolution position encoding, the proposed model can obtain multi-level global information. It overcomes the limitation of local features in locating multiple tampered regions at different scales in an image.
- Comprehensive experimental evaluation demonstrates its superior performance in detecting multi-scale tampered areas of splicing forgery images.

2 Related Work

The existing approaches for splicing images forgery detection can be generally divided into two types based on feature extraction methods: traditional ones and deep learning ones.

Traditional Ones. Traditional feature extraction-based methods mainly include image essence attributes [1], image-compression attributes [2, 11], edge inconsistencies [3, 4], and inherent attribute consistency of imaging devices [5, 6]. These methods detect image forgery based on the inconsistency of specific forgery cues. However, they suffer dramatic performance degradation when the forgery cue is missing or subject to post-processing.

Deep Learning Ones. Motivated by the tremendous success of deep learning in object detection [12, 15], some researchers have tried to use deep learning techniques to detect forged images [7, 8]. Different from object detection, forgery detection focuses only on the tampered regions. For example, Xiao et al. [7] designed a cascaded CNN network to perform a coarse-to-fine localization of tampered regions at the patch level. However, this method only focuses on local image patches and lacks of relationship between the patches, resulting in incorrect results. Thus, end-to-end-based CNN networks have been proposed for image splicing forgery detection. Bappy et al. [9] proposed to use of LSTM to extract tampering artifacts and fuse them complementarily with spatial domain features to locate manipulated regions. Wu et al. [10] designed a ManTra-Net to detect multiple forgery types by combining self-supervised learning with local anomaly detection methods. Bi et al. [13] exploited residual propagation and residual feedback mechanisms to enhance feature learning of the CNN network and effectively improve splicing image detection accuracy. However, these methods lack global information about the image and are not robust enough. Furthermore, Bi et al. [14] proposed a multi-decoder single-task network to detect tampered regions. A potential limitation of this method is that model extraction features do not distinguish their importance.

3 Proposed Method

As shown in Fig. 2, the proposed FMAT-Net consists of three parts, including an encoder branch, a transition part, and a decoder branch. First, the input images are sent to the encoder, which utilizes ResNet as the backbone to extract multi-level features F_l ($l = 1, 2, 3, 4, 5$) from the images. Second, the CSIA module in the transition part allows the interaction of different levels of information and uses an attention mechanism to enhance local features, thus facilitating the detection of forged regions. Moreover, the depth-wise convolution transformer layer (DTransformer) of the transition part can harvest the multi-level global correlations of the image. This transformer layer uses depth-wise convolution to implicitly encode positional information, allowing flexibility to handle arbitrary input sizes. Then, the multi-scale attention information output by the CSIA module and the global information extracted by the transformer layers are integrated at the corresponding resolution and fed to the decoder branch to complement the information loss. The decoder branch progressively fuses multi-level local attention and global contextual information from the transition layer for forgery region localization.

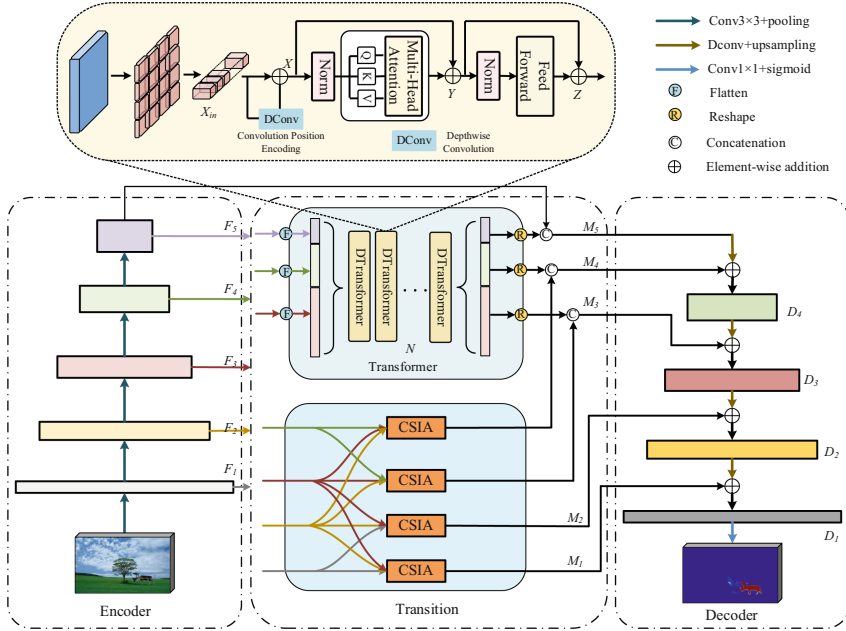


Fig. 2. The proposed model architecture diagram. The CNN encoder branch extracts features and sends them to the transition part. The transition part incorporates multi-level features by the CSIA module and extracts multi-level global contextual information by the Transformer and sends them to the decoder branch. F_l ($l = 1, 2, 3, 4, 5$) represents the l_{th} level outputs of the encoder. M_l ($l = 1, 2, 3, 4, 5$) represents the l_{th} fused features of the transition part. D_l ($l = 1, 2, 3, 4$) represents the l_{th} level outputs of the decoder stage.

3.1 Cross-Scale Interactive Attention (CSIA)

Since single-level features can only represent the information at a specific scale, thus, it is not sufficient to detect multi-scale tampered regions using single-level features. Most feature fusion methods add or concatenate different levels of feature maps directly, ignoring the interaction between different levels of features. In addition, the features extracted by CNN contain a lot of redundant information. Thus, we design a CSIA module to interactively aggregate multi-level information. Therein, the attention mechanism of CSIA adaptively selects forgery-related information from different levels of features and filters irrelevant responses. Since there is a semantic gap between different levels of features, we fuse features from adjacent levels. The architecture of the CSIA is shown in Fig. 3.

First, the features F_a and F_b from different levels of the encoder are passed through the UPConv block to achieve the same resolution and number of channels as the current level features F_c ($H \times W \times C$). $\hat{F}_a = \text{UPConv}(F_a)$, $\hat{F}_b = \text{UPConv}(F_b)$. Then, features $\hat{F}_a$, $\hat{F}_b$, and F_c are summed element by element.

$$F_{add} = \hat{F}_a \oplus \hat{F}_b \oplus F_c \quad (1)$$

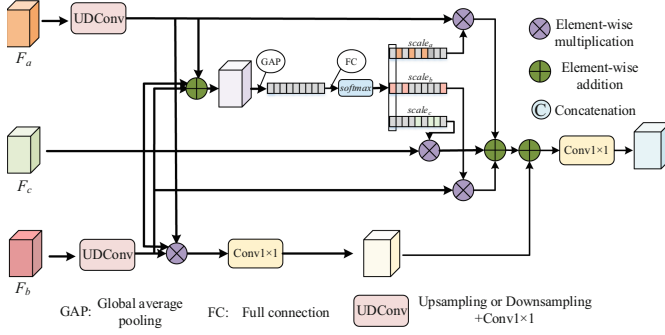


Fig. 3. The architecture of the CSIA module.

Next, the F_{add} uses the global average pooling operation (GAP) to embed spatial information and generate a $1 \times 1 \times C$ feature map, which is passed through the fully connected layers to generate a $3 \times d$ vector so that a corresponding scale vector s_i ($i = a, b, c$) can be obtained for each level features, $d = C/r$, r is a reduction ratio to control the value of the d .

$$s_i = \mathcal{F}(\mathcal{G}(F_{add})), i = a, b, c \quad (2)$$

where $\mathcal{G}(\cdot)$ denotes the global average pooling operation, $\mathcal{F}(\cdot)$ represents the fully connected (FC) layer. Then, a softmax operation is applied on the s_i channel-wise: $scale_i = \text{softmax}(s_i)$, $i = a, b, c$. Next, the features $\hat{F}_a$, $\hat{F}_b$, and F_c are multiplied by the corresponding $scale_i$ and element-wise summation of all the results to obtain the final attention information A .

$$A = \sum_{i=a,b,c} scale_i \times [\hat{F}_a, \hat{F}_b, F_c] \quad (3)$$

Furthermore, in order to fully exploit the correlations between multi-level features, we perform element-wise multiplication of different levels of features so that different levels of features can interact beneficially and promote each other. It can amplify the differences between real and forged regions and filter the noise interference in the fusion process.

$$F_{mul} = \hat{F}_a \otimes \hat{F}_b \otimes F_c \quad (4)$$

Finally, we integrate the attention features A and the interaction information F_{mul} to get the final output of the CSIA module.

3.2 Depth-Wise Convolution Transformer Layer (DTransformer)

Many existing deep learning-based splicing forgery detection methods use CNNs to extract multi-level information from images. However, due to the limited receptive field of the convolution, the features it extracts lack global relevance. To capture the global

contextual information of the image and overcome the limitation of CNNs for extracting features, we introduce a DTransformer layer. The details of the DTransformer are shown in Fig. 2.

The DTransformer layers consist of three key blocks: convolution position encoding, multi-head attention (MHA), and feed-forward network (FFN). First, deep-wise convolution (DConv) replaces the traditional position embedding method to integrate the position information into all tokens, effectively improving the flexibility of the input resolution.

$$X = DConv(X_{in}) + X_{in} \quad (5)$$

Then, we exploit the MHA to capture the global contextual relations. Finally, we add FFN to propagate the global contextual correlations.

$$Y = MHA(Norm(X)) + X \quad (6)$$

$$Z = FFN(Norm(Y)) + Y \quad (7)$$

where $Norm(\cdot)$ denotes the normalization function. In our experiments, considering the computation complexity and the performance issues, we set the number of the DTransformer layers N to 6 in all the experiments.

Table 1. Details of the datasets used for the experiments in the study.

Name	CASIA	COLUMB	NIST'16
Plain Splicing	1270	170	234
Plain Training	1220	125	184
Augmented Training	2220	725	784
Testing	50	45	50

4 Experiments

4.1 Data Set

Our experiments were run on four public datasets: Defacto [21], CASIA [16], COLUMB [17], and NIST'16 [18]. Due to we mainly focus on splicing forgery detection, we just select splicing forgery images from four datasets. Defacto is a relatively large dataset containing 105,000 computer-generated splicing images without post-processing operations. To prevent overfitting of the model, we first pre-trained the proposed model on the Defacto dataset and then fine-tuned it on the smaller datasets. For the small data sets, each dataset is randomly separated into a training set and a testing set. Then, we augment the images of each training set by random overturn, random Gaussian noise, and JPEG compression. The experimental images of three small datasets are listed in Table 1. We resize the resolution of images to 256×256 .

4.2 Evaluation Metrics

In our experiments, we use pixel-level precision, recall, and F values to evaluate the performance of the proposed FMAT-Net, which are defined as $Precision = TP/(TP + FP)$, $Recall = TP/(TP + FN)$, where TP represents the number of forged pixels predicted correctly, FP denotes the number of real pixels predicted as tampered, and FN indicates the number of altered pixels predicted as real pixels. $F = (2 \times Precision \times Recall)/(Precision + Recall)$, is the weighted harmonic mean of the precision and recall.

4.3 Implementation Details and Methods

We use the PyTorch deep learning framework to implement the proposed method. Before training, we initialize the ResNet using a pre-trained model on the ImageNet and use a stochastic gradient descent optimizer with a momentum of 0.9, and the weight decay is set to 0.0005. The proposed model is optimized by the cross-entropy loss. The batch size of 6 samples learning rate of 0.003 is set. Pre-training and fine-tuning are 100 and 50 epochs, respectively. The proposed model and all the methods used for the experiment run on an NVIDIA GeForce GTX 3090 graphics card.

To evaluate the performance of our proposed method comprehensively, we compare it with the baseline detection methods on the three public datasets. The DCT [11] and CFA [5] are the conventional methods. Their source codes are provided by the authors or reliable third parties [19]. ManTra [10], AttU-Net [20], RRU-Net [13], C2RNet [7], and RTAG [14] are deep learning-based methods. For a fair comparison, the hyper-parameters of the comparative methods are adjusted to the values that produce the best performance in the original literature.

Table 2. Results of the different detection methods on the three datasets.

	CASIA			COLUMB			NIST'16		
	P	R	F	P	R	F	P	R	F
DCT	0.346	0.864	0.494	0.387	0.684	0.494	0.124	0.817	0.215
CFA	0.108	0.916	0.193	0.546	0.468	0.504	0.164	0.876	0.276
ManTra	0.821	0.793	0.807	0.856	0.849	0.852	0.816	0.824	0.82
AttU-Net	0.816	0.787	0.801	0.836	0.796	0.816	0.781	0.776	0.778
RRU-Net	0.848	0.834	0.841	0.923	0.834	0.876	0.783	0.782	0.782
C2RNet	0.581	0.808	0.676	0.804	0.612	0.695	0.468	0.666	0.55
RTAG	–	–	0.815	–	–	0.823	–	–	0.623
Ours	0.871	0.843	0.857	0.941	0.853	0.895	0.828	0.836	0.832

4.4 Comparative Experiment and Analysis

All comparison methods are compared on the CASIA, COLUMB, and NIST'16 testing sets based on metrics such as detection precision (P), recall (R), and F-value. The detailed

results are given in Table 2, where bold entities indicate the best performance, and ‘-’ indicates that the value of the metric is not provided in the original study.

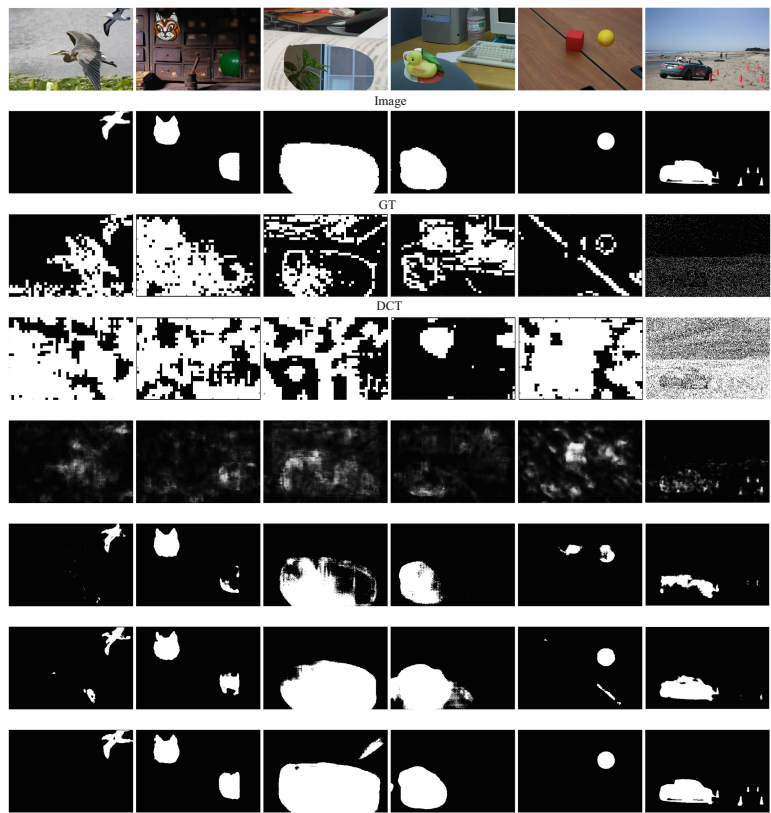


Fig. 4. Comparison of the proposed model with existing splicing forgery detection methods. 1st and 2nd columns are from CASIA, 3rd and 4th columns are from COLUMB, 5th and 6th columns are from NIST'16.

It can be seen from Table 2 that the overall performance of the proposed method performs better than other methods on all three datasets. However, the recall is a little lower than the CFA on the CASIA and NIST'16 datasets. Figure 4 illustrates the visual comparisons of the different methods. We can see from Fig. 4 that the CFA identifies many genuine regions of the image as tampered regions, so the recall is relatively high. Among the deep learning-based methods, our FMAT-Net and RRU-Net can locate most of the tampered areas. However, the proposed model can detect and locate tampered regions and authentic regions more precisely. Furthermore, the detection results of AttU-Net and ManTra are either missing or misjudged. This may be because both methods lack the long-range dependencies of the image. The comprehensive experiments demonstrate the validity of our proposed approach.

5 Conclusion

In this paper, we propose a novel method by aggregating multi-level local attention information and global contextual correlations to detect and locate the tampered regions of the splicing forgery images. Firstly, the designed CSIA module enables different levels of features to interact across scales and guides the model to pay more attention to the critical regions of the image. It enhances the learning of the local discriminative representations, improving the capacity of the model to locate multi-scale tampered regions. Secondly, a global feature mining layer based on the Transformer, dubbed DTransformer, is introduced to extract multi-level global contextual information from the forgery image to improve the detection results. Furthermore, we evaluate the performance of the proposed model and the state-of-the-art detection methods on three benchmark datasets. Extensive experiments indicate that the presented model performs better than the state-of-the-art approaches. It can effectively detect and segment multi-scale tampered regions. However, due to the introduction of the Transformer in the proposed model, its inference time will be longer. Therefore, in the future, we will consider designing more lightweight and effective models to detect splicing forged images.

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